4.3. Subspaces

20.04.2020

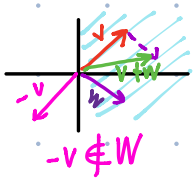
V = vector space

$W \subset V$ and W is also a v.space $\Rightarrow W$ a subspace.

THEOREM. W is a subspace if (a) $\forall \vec{v}, \vec{w} \in W \quad \vec{v}+\vec{w} \in W$

(b) $\forall c \in \mathbb{R}, \forall \vec{v} \in W, \quad c\vec{v} \in W$

ex



$-v \notin W$

$V = \mathbb{R}^2 = \left\{\begin{bmatrix} a \\ b \end{bmatrix}\right\}$, $W = \left\{\begin{bmatrix} a \\ b \end{bmatrix} : b \geq 0\right\}$

Is W a subspace of V?

No. W is not a subspace.

$v_1, v_2, \dots, v_k \in V$ and $a_1, \dots, a_k \in \mathbb{R} \Rightarrow a_1v_1 + a_2v_2 + \cdots + a_kv_k$ linear combination

ex Is $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ a lin. comb. of $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$?

$\begin{pmatrix} 1 \\ 2 \end{pmatrix} = a_1\begin{pmatrix} 2 \\ 3 \end{pmatrix} + a_2\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ has a solution a_1, a_2?

$1 = 2a_1 + 4a_2$
$2 = 3a_1 + 6a_2$ $\Rightarrow \left(\begin{array}{cc|c} 2 & 4 & 1 \\ 3 & 6 & 2 \end{array}\right) \Rightarrow -\frac{3}{2}r_1 + r_2 \to r_2 \quad \left(\begin{array}{cc|c} 2 & 4 & 1 \\ 0 & 0 & 1/2 \end{array}\right) \Rightarrow 0 \cdot a_1 + 0 \cdot a_2 = \frac{1}{2}$. No sol.

THM $A_{m \times n}$ matrix. The solution set of $A\vec{x} = \vec{0}$ is a subspace of $\mathbb{R}^n$.

proof. If $\vec{x}_1, \vec{x}_2$ are solutions $\Rightarrow A(\vec{x}_1 + \vec{x}_2) = \vec{0}$? YES.

$$A\vec{x}_1 + A\vec{x}_2 = \vec{0} + \vec{0} = \vec{0}$$

If $\vec{x}_1$ is a soln. and $c \in \mathbb{R} \Rightarrow A(c\vec{x}_1) = \vec{0}$? YES.

$$c(A\vec{x}_1) = c\vec{0} = \vec{0}$$

ex. $x_1 + x_2 = 0$, $x_2 + x_3 = 0$ $\Rightarrow$ 2 eqs. 3 unknowns. $x_3 = r$ = free, $x_2 = -r$, $x_1 = r$ $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} r \\ -r \\ r \end{bmatrix}$ $\to$ soln. set. $\searrow$ subspace of $\mathbb{R}^3$.

4.4. SPAN.

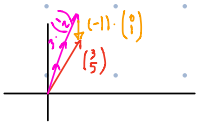
V = v.space. $v_1, v_2, \dots, v_k \in V$. The set of all possible lin. comb. of $v_1, \dots, v_k$ is called the span of $v_1, \dots, v_k$.

ex. Find the span of $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^2$.

= all lin. comb. $a_1\begin{pmatrix}1\\2\end{pmatrix}+a_2\begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}a_1\\2a_1+a_2\end{pmatrix}$. Any vector can be written like this.

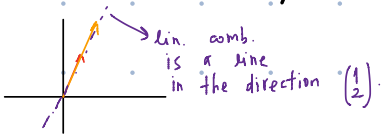
$\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}a_1\\2a_1+a_2\end{pmatrix}\Rightarrow a_1=x,\ a_2=y-2x$ ex. $\begin{pmatrix}3\\5\end{pmatrix}=3\begin{pmatrix}1\\2\end{pmatrix}+(-1)\begin{pmatrix}0\\1\end{pmatrix}$

The span of $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$ is $\mathbb{R}^2$.



ex Find the span of $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$.

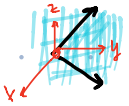
= all lin. comb. of $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$. $a_1\begin{pmatrix}1\\2\end{pmatrix}+a_2\begin{pmatrix}2\\4\end{pmatrix}=\begin{pmatrix}a_1+2a_2\\2a_1+4a_2\end{pmatrix}=(a_1+2a_2)\begin{pmatrix}1\\2\end{pmatrix}$



ex $\begin{pmatrix}1\\2\\0\end{pmatrix},\begin{pmatrix}2\\4\\0\end{pmatrix}$. Span?

→ span is a line.

ex $\begin{pmatrix}1\\2\\0\end{pmatrix},\begin{pmatrix}0\\2\\3\end{pmatrix}$



span of $\begin{pmatrix}1\\2\\0\end{pmatrix}$ and $\begin{pmatrix}0\\2\\3\end{pmatrix}$ is $a_1\begin{pmatrix}1\\2\\0\end{pmatrix}+a_2\begin{pmatrix}0\\2\\3\end{pmatrix}=\begin{pmatrix}a_1\\2a_1+2a_2\\3a_2\end{pmatrix}$

span is all vectors of this form.

$\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}a_1\\2a_1+2a_2\\3a_2\end{pmatrix}\Rightarrow a_1=x,\ a_2=\frac{z}{3},\ y=2x+\frac{2z}{3}$

$\begin{pmatrix}1\\0\\3\end{pmatrix}$ is in the span?

$a_1=1,\ a_2=1$ $0\neq 2+2$. No, it is not.

ex $V=M_{2\times 2}$ $\vec{v_1}=\begin{pmatrix}1&0\\0&0\end{pmatrix},\ \vec{v_2}=\begin{pmatrix}0&1\\0&0\end{pmatrix},\ \vec{v_3}=\begin{pmatrix}0&0\\1&0\end{pmatrix}$

Span of v_1,v_2,v_3 → $\begin{pmatrix}a&b\\c&0\end{pmatrix}=a\vec{v_1}+b\vec{v_2}+c\vec{v_3}$

THM. Span $\{v_1,\dots,v_k\}$ is a subspace of V.

ex $V=P_2$ = v.space of polynomials with degree ≤ 2.

$S=\{t^2,t\}\Rightarrow$ span S = all polys of the form $at+bt^2$.

ex Is t^2-t in the span of t^2+1 and $t-1$?

$t^2-t=a(t^2+1)+b(t-1)=t^2a+tb+(a-b)\Rightarrow a=1,\ b=-1,\ a-b=0$. No soln.

ex $v_1=\begin{bmatrix}2\\1\\1\end{bmatrix},\ v_2=\begin{bmatrix}1\\-1\\3\end{bmatrix}$ $v=\begin{bmatrix}1\\5\\-7\end{bmatrix}$. Is v in the span of v_1,v_2?

Is there a soln. of $v=av_1+bv_2$ for a,b?

$\begin{bmatrix}1\\5\\-7\end{bmatrix}=a\begin{bmatrix}2\\1\\1\end{bmatrix}+b\begin{bmatrix}1\\-1\\3\end{bmatrix}\Rightarrow\begin{matrix}2a+b=1\\a-b=5\\a+3b=-7\end{matrix}\Rightarrow\left[\begin{array}{cc|c}2&1&1\\1&-1&5\\1&3&-7\end{array}\right]\xrightarrow{r_1\leftrightarrow r_2}\left[\begin{array}{cc|c}1&-1&5\\2&1&1\\1&3&-7\end{array}\right]\begin{matrix}-2r_1+r_2\to r_2\\-r_1+r_3\to r_3\end{matrix}\left[\begin{array}{cc|c}1&-1&5\\0&3&-9\\0&4&-12\end{array}\right]\begin{matrix}\frac{r_2}{3}\to r_2\\ \\ r_3/4\to r_3\end{matrix}\left[\begin{array}{cc|c}1&-1&5\\0&1&-3\\0&1&-3\end{array}\right]-r_2+r_3\to r_3$

YES.

$\left[\begin{array}{cc|c}1&-1&5\\0&1&-3\\0&0&0\end{array}\right]$ ✓ $b=-3$, $a-b=5$, $a=2$.

ex $v_1=\begin{bmatrix}1\\2\\1\end{bmatrix},\ v_2=\begin{bmatrix}1\\0\\2\end{bmatrix},\ v_3=\begin{bmatrix}1\\1\\0\end{bmatrix}$. Do these vectors span $\mathbb{R}^3$?

Can I write $\begin{bmatrix}x\\y\\z\end{bmatrix}=av_1+bv_2+cv_3$? $\begin{matrix}x=a+b+c\\y=2a+0b+c\\z=a+2b+0c\end{matrix}$. Can I solve for a,b,c in terms of x,y,z?

$\left[\begin{array}{ccc|c}1&1&1&x\\2&0&1&y\\1&2&0&z\end{array}\right]\xrightarrow{-2r_1+r_2\to r_2,\ -r_1+r_3\to r_3}\left[\begin{array}{ccc|c}1&1&1&x\\0&-2&-1&y-2x\\0&1&-1&z-x\end{array}\right]\xrightarrow{r_2\leftrightarrow r_3}\left[\begin{array}{ccc|c}1&1&1&x\\0&1&-1&z-x\\0&-2&-1&y-2x\end{array}\right]\xrightarrow{2r_2+r_3\to r_3}\left[\begin{array}{ccc|c}1&1&1&x\\0&1&-1&z-x\\0&0&-3&y-2x+2(z-x)\end{array}\right]$ $c=\frac{y-2x+2(z-x)}{-3}$

The answer is yes.

ex $V=P_2$. $v_1=t^2+2t+1$, $v_2=t^2+2$. Does v_1,v_2 span P_2?

$a+bt+ct^2=a_1v_1+a_2v_2$. Can we solve for a_1,a_2?

$=a_1(t^2+2t+1)+a_2(t^2+2)=t^2(a_1+a_2)+t(2a_1)+a_1+2a_2$

$\begin{array}{l}a_1+a_2=c\\2a_1=b\\a_1+2a_2=a\end{array}\Rightarrow\begin{array}{l}a_1=b/2\\a_2=c-b/2\end{array}\Rightarrow\boxed{\frac{b}{2}+2\left(c-\frac{b}{2}\right)=a}\Rightarrow 2c-\frac{b}{2}=a$

For example $a=1, b=2, c=1\Rightarrow 1+2t+t^2$ is in the span.

But $a=1, b=2, c=4\Rightarrow 1+2t+4t^2$ is not in the span. The answer is NO!

21.04.2020

ex $A=\begin{bmatrix}1&1&0&2\\-2&-2&1&-5\\1&1&-1&3\\4&4&-1&9\end{bmatrix}$ $A\vec{x}=\vec{0}$. Find a set of vectors which span the solution space.

$\left[\begin{array}{c|c}A&\begin{matrix}0\\0\\0\\0\end{matrix}\end{array}\right]\xrightarrow{\begin{array}{l}2r_1+r_2\to r_2\\-r_1+r_3\to r_3\\-4r_1+r_4\to r_4\end{array}}\left[\begin{array}{cccc|c}1&1&0&2&0\\0&0&1&-1&0\\0&0&-1&1&0\\0&0&-1&1&0\end{array}\right]\xrightarrow{\begin{array}{l}r_2+r_3\to r_3\\r_2+r_4\to r_4\end{array}}\left[\begin{array}{cccc|c}x_1&x_2&x_3&x_4&\\1&1&0&2&0\\0&0&1&-1&0\\0&0&0&0&0\\0&0&0&0&0\end{array}\right]$

$x_1+x_2+2x_4=0$, $x_3-x_4=0$, $x_2=r$ (free), $x_4=s$, $x_3=s$, $x_1=-r-2s$

$\vec{x}=\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}=\begin{bmatrix}-r-2s\\r\\s\\s\end{bmatrix}=\begin{bmatrix}-r\\r\\0\\0\end{bmatrix}+\begin{bmatrix}-2s\\0\\s\\s\end{bmatrix}=r\begin{bmatrix}-1\\1\\0\\0\end{bmatrix}+s\begin{bmatrix}-2\\0\\1\\1\end{bmatrix}$ Sol. Sp. $=\operatorname{span}\left\{\begin{bmatrix}-1\\1\\0\\0\end{bmatrix},\begin{bmatrix}-2\\0\\1\\1\end{bmatrix}\right\}$

4.5. LINEAR INDEPENDENCE

$W=\left\{\begin{bmatrix}a\\b\\a+b\end{bmatrix}\right\}\subseteq\mathbb{R}^3$ subspace.

$W=\operatorname{span}\left\{\begin{bmatrix}1\\0\\1\end{bmatrix},\begin{bmatrix}0\\1\\1\end{bmatrix}\right\}=\operatorname{span}\left\{\begin{bmatrix}1\\0\\1\end{bmatrix},\begin{bmatrix}0\\1\\1\end{bmatrix},\begin{bmatrix}1\\1\\2\end{bmatrix}\right\}$

V= v. space, $\vec{v}_1,\dots,\vec{v}_k\in V$ linearly dependent if there exists scalars $a_1,\dots,a_k$ not all zero

$a_1\vec{v}_1+a_2\vec{v}_2+\dots+a_k\vec{v}_k=\vec{0}$

Nontrivial sol.

If $\{v_1,\dots,v_k\}$ is not lin. dep. $\Rightarrow$ lin. independent.

only trivial soln.

ex Is $\left\{\begin{bmatrix}1\\0\\1\end{bmatrix},\begin{bmatrix}0\\1\\1\end{bmatrix}\right\}$ lin. dep. or indep.?

sol. $a_1\begin{bmatrix}1\\0\\1\end{bmatrix}+a_2\begin{bmatrix}0\\1\\1\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}\Rightarrow\begin{array}{l}a_1=0\\a_2=0\\a_1+a_2=0\end{array}$. Only 1 sol. $a_1=a_2=0$. Trivial sol. $\Rightarrow$ lin. indep.

ex Is $\left\{\begin{bmatrix}1\\0\\1\end{bmatrix},\begin{bmatrix}0\\1\\1\end{bmatrix},\begin{bmatrix}1\\1\\2\end{bmatrix}\right\}$ lin. dep. or indep.?

sol. $a_1\begin{bmatrix}1\\0\\1\end{bmatrix}+a_2\begin{bmatrix}0\\1\\1\end{bmatrix}+a_3\begin{bmatrix}1\\1\\2\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix} \Rightarrow \begin{matrix}a_1+a_3=0\\a_2+a_3=0\\a_1+a_2+2a_3=0\end{matrix}$ $\left[\begin{array}{ccc|c}1&0&1&0\\0&1&1&0\\1&1&2&0\end{array}\right] \Rightarrow \left[\begin{array}{ccc|c}1&0&1&0\\0&1&1&0\\0&1&1&0\end{array}\right]_{-2} \Rightarrow \left[\begin{array}{ccc|c}1&0&1&0\\0&1&1&0\\0&0&0&0\end{array}\right]$

$a_3=r=\text{free}$, $a_2=-r$, $a_1=-r$ $\Rightarrow \begin{bmatrix}a_1\\a_2\\a_3\end{bmatrix}=\begin{bmatrix}-r\\-r\\r\end{bmatrix}$ $\Rightarrow$ Nontrivial solns $\Rightarrow$ lin. dep.

27.04.2020

ex. $v_1=[1\ 0\ 0]$, $v_2=[0\ 1\ 0]$ $v_3=[0\ 0\ 1]$. Are these vectors lin. dep. or indep?

sol. $a_1\vec{v}_1+a_2\vec{v}_2+a_3\vec{v}_3=\vec{0} \Rightarrow a_1[1\ 0\ 0]+a_2[0\ 1\ 0]+a_3[0\ 0\ 1]=[0\ 0\ 0]$

$[a_1\ \ a_2\ \ a_3]=[0\ 0\ 0] \Rightarrow a_1=a_2=a_3=0$. Lin. indep.

ex. $V=P_2$. $v_1=t^2+t+2$, $v_2=2t^2+t$, $v_3=3t^2+2t+2$. Lin. dep or lin. indep?

sol. $a_1(t^2+t+2)+a_2(2t^2+t)+a_3(3t^2+2t+2)=0$

1st method: $t^2(a_1+2a_2+3a_3)+t(a_1+a_2+2a_3)+(2a_1+2a_3)=0$ (coefficients $\to 0$, $\to 0$, $\to 0$)

Recall $A_{n\times n}$, $A\vec{x}=\vec{0}$: $\det A=0$ $\Rightarrow$ nontrivial sol.; $\det A\neq 0$ $\Rightarrow$ only the trivial solution.

$\begin{matrix}a_1+2a_2+3a_3=0\\a_1+a_2+2a_3=0\\2a_1+2a_3=0\end{matrix} \Rightarrow \begin{vmatrix}1&2&3\\1&1&2\\2&0&2\end{vmatrix}=(-1)^{3+1}2\begin{vmatrix}2&3\\1&2\end{vmatrix}+0+(-1)^{3+3}2\begin{vmatrix}1&2\\1&1\end{vmatrix}=2\cdot(4-3)+2(1-2)=2-2=0.$

Nontrivial sol. $\Rightarrow$ lin. dep.

2nd method. $v_1+v_2=v_3 \Rightarrow 1v_1+1v_2-1v_3=0 \Rightarrow$ lin. dep.

thm. $v_1,\dots,v_k$ in $\mathbb{R}^k$ are lin. indep. $\Leftrightarrow \det([v_1\cdots v_k])\neq 0$

lin. dep. $\Leftrightarrow \det([v_1\cdots v_k])=0$

ex. $S=\left\{\begin{bmatrix}1\\2\\3\end{bmatrix},\begin{bmatrix}0\\1\\2\end{bmatrix},\begin{bmatrix}3\\0\\-1\end{bmatrix}\right\}$. Lin. dep. or lin. indep.?

sol. Check $\begin{vmatrix}1&0&3\\2&1&0\\3&2&-1\end{vmatrix}=2 \Rightarrow$ lin. indep.

thm. If at least one of $v_1,\dots,v_k$ is the zero-vec. then $v_1,\dots,v_k$ is lin. dep.

thm. $V=$ v. space. $S_1\subset S_2\subset V$. If S_1 is lin. dep. $\Rightarrow S_2$ is lin. dep.

If S_2 is lin. indep. $\Rightarrow S_1$ is lin. indep.

ex. v_1,v_2,v_3 is lin. dep. $\Rightarrow v_1,v_2,v_3,v_4$ is also lin. dep.

Geometric idea

4.6. BASIS AND DIMENSION

def. $V=$ v. space. $v_1,\dots,v_k$ in V forms a **basis** for V if:

a) $v_1,\dots,v_k$ lin. indep. b) span $\{v_1,\dots,v_k\}=V$.

Remark. If $v_1, \dots, v_k$ form a basis $\Rightarrow$ they must be distinct and non-zero.

ex. For $\mathbb{R}^2$ $\left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$ form a standard basis.

a) Lin. indep. $a_1\begin{bmatrix}1\\0\end{bmatrix}+a_2\begin{bmatrix}0\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix} \Rightarrow a_1=a_2=0 \Rightarrow$ lin. indep.

b) $\operatorname{span}\left(\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right)=\mathbb{R}^2 \Rightarrow \begin{bmatrix}x\\y\end{bmatrix}= x\begin{bmatrix}1\\0\end{bmatrix}+y\begin{bmatrix}0\\1\end{bmatrix}$

ex For $\mathbb{R}^2$ $\left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}$ form a basis for $\mathbb{R}^2$.

a) Lin. indep $\Rightarrow \begin{vmatrix}1&1\\0&1\end{vmatrix}=1\neq 0$

b) $\begin{bmatrix}x\\y\end{bmatrix}= a\begin{bmatrix}1\\0\end{bmatrix}+ b\begin{bmatrix}1\\1\end{bmatrix}$ $\quad x=a+b,\ y=0+b \Rightarrow b=y,\ a=x-b=x-y.$

ex. For $\mathbb{R}^3$ $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ form a standard basis.

ex $V=P_2$. $S=\{t^2+1,\ t-1,\ 2t+2\}$. Is S a basis for P_2?

sol a) lin. indep. $a(t^2+1)+b(t-1)+c(2t+2)=0$

$$at^2+(b+2c)t+(a-b+2c)=0 \rightarrow \left.\begin{matrix}a=0\\b+2c=0\\a-b+2c=0\end{matrix}\right\} \Rightarrow \begin{matrix}c=0\\b=0\end{matrix}$$

only Trivial Soln. $\Rightarrow$ Lin. indep.

b) span $S=P_2$? YES!

$$at^2+bt+c = a_1(t^2+1)+a_2(t-1)+a_3(2t+2)$$

$$\left.\begin{matrix}a_1=a\\a_2+2a_3=b\\a_1-a_2+2a_3=c\end{matrix}\right\} \Rightarrow \begin{matrix}a_2+2a_3=b\\-a_2+2a_3=c-a\end{matrix} \Rightarrow 4a_3=b+c-a,\ 2a_2=b-c+a$$

$$\begin{bmatrix}1&0&0\\0&1&2\\1&-1&2\end{bmatrix}\begin{bmatrix}a_1\\a_2\\a_3\end{bmatrix}=\begin{bmatrix}a\\b\\c\end{bmatrix}$$

Recall. $A_{n\times n}$ $A\vec{x}=\vec{b}$. If A^{-1} exists $\vec{x}=A^{-1}\vec{b}$.

$$\updownarrow$$

$$\det A\neq 0$$

For the prev. problem. It is sufficient to check $\det\left(\begin{bmatrix}1&0&0\\0&1&2\\1&-1&2\end{bmatrix}\right)\neq 0 \Rightarrow$ Basis!

ex. $V=P_2$. $W\subset V$, $W=\{at^2+bt+(a-b)\}$ subspace V. ex $2t^2+3t-1 \in W$, $3t^2+1 \notin W$

Find a basis for W?

sol $at^2+bt+(a-b)=a(t^2+1)+b(t-1)$. Choose $S=\{t^2+1, t-1\}$. Show S is a basis for W.

a) lin. indep. $a(t^2+1)+b(t-1)=0 \Rightarrow \begin{matrix}a=0\\b=0\end{matrix}$ only trivial solution $\Rightarrow$ lin. indep.

b) span$(S)=W$? ✓.

def. $V=$ v.space. and let S be a basis of V.

if S has ∞ vectors $\Rightarrow V=\infty$ dimensional v. space.

S has finitely many vectors $\Rightarrow$ V = finite dim. v.space.

Number of vectors in S is called the **dimension** of V.

Axiom of choice $\Rightarrow$ Any v.space has a basis.

The number of vectors in S is independent of S.

ex $v_1=\begin{bmatrix}1\\0\\1\end{bmatrix}$, $v_2=\begin{bmatrix}0\\1\\1\end{bmatrix}$, $v_3=\begin{bmatrix}1\\1\\2\end{bmatrix}$, $v_4=\begin{bmatrix}1\\2\\1\end{bmatrix}$, $v_5=\begin{bmatrix}-1\\1\\-2\end{bmatrix}$

$S=\{v_1,\dots,v_5\}$. Find a basis for span(S).

sol. $a_1\vec{v}_1+\cdots+a_5\vec{v}_5=\begin{bmatrix}0\\0\\0\end{bmatrix} \Rightarrow \left[\begin{array}{ccccc|c}1&0&1&1&-1&0\\0&1&1&2&1&0\\1&1&2&1&-2&0\end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccccc|c}1&0&1&0&-2&0\\0&1&1&0&-1&0\\0&0&0&1&1&0\end{array}\right]$ $\begin{matrix}a_3=r\\a_5=s\\a_4=-s\end{matrix}$ $\begin{matrix}a_1=-r+2s\\a_2=-r+s\end{matrix}$

$\text{span}(S)=\underbrace{\{v_1,v_2,v_4\}}_{\text{span}}$ → basis ↳ We choose vectors which have a leading 1.

Suppose we remove $v_3, v_5 \Rightarrow$ RREF $\left[\begin{array}{ccc|c}1&0&0&0\\0&1&0&0\\0&0&1&0\end{array}\right]$

28.04

(cont.) $(-r+2s)\vec{v}_1+(-r+s)\vec{v}_2+r\vec{v}_3-s\vec{v}_4+s\vec{v}_5=\vec{0}$

$r(-\vec{v}_1-\vec{v}_2+\vec{v}_3)+s(2\vec{v}_1+\vec{v}_2-\vec{v}_4+\vec{v}_5)=\vec{0}$ for every r,s.

$s=0,\ r=1 \Rightarrow -\vec{v}_1-\vec{v}_2+\vec{v}_3=\vec{0} \Rightarrow \vec{v}_3=\vec{v}_1+\vec{v}_2$

$r=0,\ s=1 \Rightarrow \vec{v}_5=-2\vec{v}_1-\vec{v}_2+\vec{v}_4$

thm. If $\vec{v}=a_1\vec{v}_1+\cdots+a_k\vec{v}_k$, then $\text{span}\{\vec{v},\vec{v}_1,\dots,\vec{v}_k\}=\text{span}\{\vec{v}_1,\dots,\vec{v}_k\}$.

ex. $V=\mathbb{R}_3$. $S=\{\vec{v}_1,\vec{v}_2,\vec{v}_3\}$. $\vec{v}_1=[0\ 1\ 1]$, $\vec{v}_2=[1\ 0\ 1]$ $\vec{v}_3=[1\ 1\ 2]$

Find a basis for spanS. Find $\dim(\text{span}\{\vec{v}_1,\vec{v}_2,\vec{v}_3\})$

sol. Observe: $\vec{v}_3=\vec{v}_1+\vec{v}_2$. $\text{span}\{\vec{v}_1,\vec{v}_2,\vec{v}_3\}=\text{span}\{\vec{v}_1,\vec{v}_2\}$

We are done if $\{\vec{v}_1,\vec{v}_2\}$ is lin. indep.

$a_1\vec{v}_1+a_2\vec{v}_2=\vec{0} \Rightarrow a_1[0\ 1\ 1]+a_2[1\ 0\ 1]=[0\ 0\ 0] \Rightarrow [a_2,a_1,a_1+a_2]=[0\ 0\ 0]$

$a_1=a_2=0$. Only the trivial sol. exists.

So $\{\vec{v}_1,\vec{v}_2\}$ are both lin. indep. and span $\text{span}\{\vec{v}_1,\vec{v}_2,\vec{v}_3\}$. So $\{\vec{v}_1,\vec{v}_2\}$ is a basis for $\text{span}\{\vec{v}_1,\vec{v}_2,\vec{v}_3\}$.

$\dim(\text{span}\{\vec{v}_1,\vec{v}_2,\vec{v}_3\})=2$.

thm. $\dim V=n$. $\Rightarrow$ any subset S of V with more than n vectors must be lin. dep.

$\Rightarrow$ any subset ... with fewer than ...

can not span V.

ex Find a basis for P_3.

sol $a_3t^3+a_2t^2+a_1t+a_0 1 \Rightarrow P_3 = \text{span}\{1, t, t^2, t^3\}$. Moreover, they are lin. indep.

$a_0+a_1t+a_2t^2+a_3t^3=0 \Leftrightarrow a_0=a_1=a_2=a_3=0$

$\dim(P_3)=4$. $\dim(P_n)=n+1$.

ex Is $\{t^2-1, t+1, t-1, t^2+t\}$ a basis for P_2?

sol No. More than 3 vectors in P_2 must be lin. dep.

4.7. Homogeneous Systems.

$A_{m\times n}$ matrix. $A\vec{x}=\vec{0}$ $\Leftrightarrow$ Homogeneous sys.

thm The solution set of $A\vec{x}=\vec{0}$ is a subspace of $\mathbb{R}^n$.

proof If $\vec{x}_1$ and $\vec{x}_2$ are solutions, $A\vec{x}_1=\vec{0}$, $A\vec{x}_2=\vec{0} \Rightarrow A(\vec{x}_1+\vec{x}_2)=A\vec{x}_1+A\vec{x}_2=\vec{0}+\vec{0}=\vec{0}$.

$c\in\mathbb{R}$, $\vec{x}$ is a soln. $\Rightarrow A(c\vec{x})=cA\vec{x}=c\vec{0}=\vec{0}$.

def Sol. space of $A\vec{x}=\vec{0}$ is called nullspace of A and dimension of nullspace is called nullity.

ex Suppose that RREF of $A\vec{x}=\vec{0}$ $\left[\begin{array}{ccccc|c} 1&0&2&0&1&0\\ 0&1&2&0&-1&0\\ 0&0&0&1&2&0\\ 0&0&0&0&0&0 \end{array}\right]$ Find a basis for the sol. sp.

$\vec{x}=\begin{bmatrix} x_1\\ x_2\\ x_3\\ x_4\\ x_5 \end{bmatrix}$ $x_3=r$, $x_5=s$; $x_4+2x_5=0$, $x_4=-2x_5=-2s$; $x_1=-2x_3-x_5=-2r-s$; $x_2=-2x_3+x_5=-2r+s$

$\vec{x}=\begin{bmatrix} -2r-s\\ -2r+s\\ r\\ -2s\\ s \end{bmatrix}=r\begin{bmatrix} -2\\ -2\\ 1\\ 0\\ 0 \end{bmatrix}+s\begin{bmatrix} -1\\ 1\\ 0\\ -2\\ 1 \end{bmatrix}$. So $\begin{bmatrix} -2\\ -2\\ 1\\ 0\\ 0 \end{bmatrix}, \begin{bmatrix} -1\\ 1\\ 0\\ -2\\ 1 \end{bmatrix}$ form a basis for the sol. sp.

Nullity of $A=2$.

remark The solution set of $A\vec{x}=\vec{b}$, $\vec{b}\neq\vec{0}$ is not a vector space. Because if $A\vec{x}_1=\vec{b}$, $A\vec{x}_2=\vec{b} \Rightarrow A(\vec{x}_1+\vec{x}_2)=2\vec{b}\neq\vec{b}$

Any solution of $A\vec{x}=\vec{b}$ can be written as a sum of a soln. of $A\vec{x}=\vec{0}$ and one particular soln. of $A\vec{x}=\vec{b}$.

4/05

4.8. COORDINATES.

V = v.space. An ordered basis $S = \{\vec{v}_1, \cdots, \vec{v}_n\}$.

$\vec{v} \in V \Rightarrow \vec{v} = a_1\vec{v}_1 + \cdots + a_n\vec{v}_n$, $a_1, \cdots, a_n \in \mathbb{R}$.

$[\vec{v}]_S = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$ = coordinates of $\vec{v}$ w.r.t. S.

ex $V = \mathbb{R}^2$. $S = \left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}1\\1\end{bmatrix}\right\}$ basis.

$\vec{v} = \begin{bmatrix}3\\2\end{bmatrix}$. $[\vec{v}]_S = ?$

solution. $\begin{bmatrix}3\\2\end{bmatrix} = a_1\begin{bmatrix}1\\0\end{bmatrix} + a_2\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}a_1+a_2\\a_2\end{bmatrix} \Rightarrow \begin{matrix}a_2=2\\a_1=1\end{matrix} \Rightarrow [\vec{v}]_S = \begin{bmatrix}1\\2\end{bmatrix}$. To obtain $\begin{bmatrix}3\\2\end{bmatrix}$,

ex $S = \left\{\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\}$. $\begin{bmatrix}3\\2\end{bmatrix}_S = \begin{bmatrix}2\\1\end{bmatrix}$

$S = \{v_1, \cdots, v_n\}$.

$[\vec{v}]_S = \begin{bmatrix}a_1\\\vdots\\a_n\end{bmatrix}$, $[\vec{w}]_S = \begin{bmatrix}b_1\\\vdots\\b_n\end{bmatrix}$. $[\vec{v}+\vec{w}]_S = \begin{bmatrix}a_1+b_1\\\vdots\\a_n+b_n\end{bmatrix}$. Why?

$\vec{v} = a_1\vec{v}_1 + \cdots + a_n\vec{v}_n$, $\vec{w} = b_1\vec{v}_1 + \cdots + b_n\vec{v}_n$. $\vec{v}+\vec{w} = (a_1+b_1)\vec{v}_1 + \cdots + (a_n+b_n)\vec{v}_n$

$c \in \mathbb{R}$ $[c\vec{v}]_S = c[\vec{v}]_S$. proof similar.

Transition Matrices.

$S = \{\vec{v}_1, \cdots, \vec{v}_n\}$, $T = \{\vec{w}_1, \cdots, \vec{w}_n\}$ ordered bases

$\vec{v} = c_1\vec{w}_1 + \cdots + c_n\vec{w}_n \Rightarrow [\vec{v}]_T = \begin{bmatrix}c_1\\\vdots\\c_n\end{bmatrix}$

$[\vec{v}]_S = [c_1\vec{w}_1 + \cdots + c_n\vec{w}_n]_S = c_1[\vec{w}_1]_S + \cdots + c_n[\vec{w}_n]_S = \begin{bmatrix}[w_1]_S & [w_2]_S & \cdots & [w_n]_S\end{bmatrix}\begin{bmatrix}c_1\\c_2\\\vdots\\c_n\end{bmatrix}$.

$= P_{S \leftarrow T}[v]_T$

↳ transition matrix from the T-basis to S-basis.

Reminder

ex $A = \begin{bmatrix}a & b\\c & d\end{bmatrix}$ $\vec{x} = \begin{bmatrix}y\\z\end{bmatrix}$

$A\vec{x} = \begin{bmatrix}ay+bz\\cy+dz\end{bmatrix} = y\begin{bmatrix}a\\c\end{bmatrix} + z\begin{bmatrix}b\\d\end{bmatrix}$

ex $V = \mathbb{R}^2$. $S = \left\{\underbrace{\begin{bmatrix}1\\0\end{bmatrix}}_{\vec{v}_1}, \underbrace{\begin{bmatrix}1\\1\end{bmatrix}}_{\vec{v}_2}\right\}$. $T = \left\{\underbrace{\begin{bmatrix}0\\1\end{bmatrix}}_{\vec{w}_1}, \underbrace{\begin{bmatrix}-1\\0\end{bmatrix}}_{\vec{w}_2}\right\}$ (a) $P_{S \leftarrow T} = ?$

$\vec{w}_1 = c_1\vec{v}_1 + c_2\vec{v}_2 \Rightarrow \begin{bmatrix}0\\1\end{bmatrix} = c_1\begin{bmatrix}1\\0\end{bmatrix} + c_2\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}c_1+c_2\\c_2\end{bmatrix} \Rightarrow \begin{matrix}c_2=1\\c_1=-1\end{matrix}$ $[\vec{w}_1]_S = \begin{bmatrix}-1\\1\end{bmatrix}$

$\vec{w}_2 = c_1\vec{v}_1 + c_2\vec{v}_2 \Rightarrow \begin{bmatrix}-1\\0\end{bmatrix} = c_1\begin{bmatrix}1\\0\end{bmatrix} + c_2\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}c_1+c_2\\c_2\end{bmatrix} \Rightarrow \begin{matrix}c_2=0\\c_1=-1\end{matrix} \Rightarrow [\vec{w}_2]_S = \begin{bmatrix}-1\\0\end{bmatrix}$

$P_{S \leftarrow T} = \begin{bmatrix}-1 & -1\\1 & 0\end{bmatrix}$.

(b) If $[v]_T = \begin{bmatrix}2\\3\end{bmatrix} \Rightarrow [v]_S = ?$

$[\vec{v}]_S = P_{S \leftarrow T}[\vec{v}]_T = \begin{bmatrix}-1 & 1\\1 & 0\end{bmatrix}\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}1\\2\end{bmatrix}$

ex/ $V = P_1$ = polys of deg ≤ 1.

$\vec{v}_1 = t$, $\vec{v}_2 = t-3$, $\vec{w}_1 = t-1$, $\vec{w}_2 = t+1$.

$S = \{\vec{v}_1, \vec{v}_2\}$, $T = \{\vec{w}_1, \vec{w}_2\}$.

a) Find $P_{S \leftarrow T}$

$[w_1]_S = ?$ $\Leftrightarrow \vec{w}_2 = c_1\vec{v}_1 + c_2\vec{v}_2 \Leftrightarrow t-1 = c_1 t + c_2(t-3) = t(c_1+c_2) - 3c_2 \Rightarrow c_2 = 1/3 \quad c_1 = 2/3$

$[w_1]_S = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}$

$[w_2]_S = ? \Leftrightarrow t+1 = c_1 t + c_2(t-3) = t(c_1+c_2) - 3c_2 \Rightarrow \begin{matrix} c_2 = -1/3 \\ c_1 = 4/3 \end{matrix}$

$[w_2]_S = \begin{bmatrix} 4/3 \\ -1/3 \end{bmatrix}$

$P_{S \leftarrow T} = \begin{bmatrix} [w_1]_S & [w_2]_S \end{bmatrix} = \begin{bmatrix} 2/3 & 4/3 \\ 1/3 & -1/3 \end{bmatrix}$

(b) $v = 5t+1$. Find $[\vec{v}]_T$, $[\vec{v}]_S$.

$5t+1 = c_1 w_1 + c_2 w_2 = c_1(t-1) + c_2(t+1) \Rightarrow \begin{matrix} c_1+c_2 = 5 \\ c_2 - c_1 = 1 \end{matrix} \Rightarrow \begin{matrix} c_2 = 3 \\ c_1 = 2 \end{matrix} \Rightarrow [5t+1]_T = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$5t+1 = c_1 v_1 + c_2 v_2 = c_1 t + c_2(t-3) \Rightarrow \begin{matrix} c_1+c_2 = 5 \\ -3c_2 = 1 \end{matrix} \Rightarrow \begin{matrix} c_2 = -1/3 \\ c_1 = 16/3 \end{matrix} \Rightarrow [5t+1]_S = \begin{bmatrix} 16/3 \\ -1/3 \end{bmatrix}$

(c) Verify $[\vec{v}]_S = P_{S \leftarrow T}[\vec{v}]_T$. $\quad [\vec{v}]_S = \begin{bmatrix} 2/3 & 4/3 \\ 1/3 & -1/3 \end{bmatrix}\begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4/3 + 12/3 \\ 2/3 - 3/3 \end{bmatrix} = \begin{bmatrix} 16/3 \\ -1/3 \end{bmatrix}$ ✓

$P_{T \leftarrow S}$ = transition matrix from $S \to T$

$P_{S \leftarrow T}$ = trans. mat. from $T \to S$

$P_{T \leftarrow S} = (P_{S \leftarrow T})^{-1}$

Read: 4.9. Rank of a matrix.

CHAPTER 6.

6.1. Linear Transformations.

V, W = v. spaces. $L: V \to W$ is called a **linear transformation** from V to W if

(a) $L(\vec{u}+\vec{v}) = L(\vec{u}) + L(\vec{v})$, $\forall \vec{u}, \vec{v} \in V$

(b) $L(c\vec{u}) = cL(\vec{u})$, $\forall c \in \mathbb{R}$, $\forall \vec{u} \in V$

ex/ $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2\times 2}$ $\quad L: \mathbb{R}^2 \to \mathbb{R}^2$ $\quad L(\vec{u}) = A\vec{u}$. Is L a lin. trans.?

Yes. $L(\vec{u}+\vec{v}) = A(\vec{u}+\vec{v}) = A\vec{u} + A\vec{v} = L(\vec{u}) + L(\vec{v})$ ✓

$L(c\vec{u}) = A(c\vec{u}) = cA\vec{u} = cL(\vec{u})$ ✓

ex $A_{m\times n}$ matrix $L(\vec{u}) = A\vec{u}$, $L: \mathbb{R}^n \to \mathbb{R}^m$ is a lin. trans.

ex $L: P_2 \to \mathbb{R}^2$, $L(a_0+a_1t+a_2t^2) = \begin{bmatrix} a_1 \\ a_0+a_2 \end{bmatrix}$. Is this a lin. trans.? YES!

a) $L(a_0+a_1t+a_2t^2 + b_0+b_1t+b_2t^2) = L((a_0+b_0)+(a_1+b_1)t+(a_2+b_2)t^2) = \begin{bmatrix} a_1+b_1 \\ a_0+b_0+a_2+b_2 \end{bmatrix}$

$L(a_0+a_1t+a_2t^2)+L(b_0+b_1t+b_2t^2) = \begin{bmatrix} a_1 \\ a_0+a_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_0+b_2 \end{bmatrix}$ ← YES!

b) $L(c(a_0+a_1t+a_2t^2)) = L(ca_0+ca_1t+ca_2t^2) = \begin{bmatrix} ca_1 \\ ca_0+ca_2 \end{bmatrix}$

$cL(a_0+a_1t+a_2t^2) = c\begin{bmatrix} a_1 \\ a_0+a_2 \end{bmatrix}$ ← YES!

ex $L: \mathbb{R}^3 \to \mathbb{R}^2$. $L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} x+1 \\ 2y \end{bmatrix}$. Is L a lin. trans? NO!

a) $L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) + L\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = L\left(\begin{bmatrix} x+a \\ y+b \\ z+c \end{bmatrix}\right) = \begin{bmatrix} x+a+1 \\ 2(y+b) \end{bmatrix}$ ← not equal.

$L\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) + L\left(\begin{bmatrix} a \\ b \\ c \end{bmatrix}\right) = \begin{bmatrix} x+1 \\ 2y \end{bmatrix} + \begin{bmatrix} a+1 \\ 2b \end{bmatrix} = \begin{bmatrix} x+a+2 \\ 2y+2b \end{bmatrix}$

$L(\vec{u}+\vec{v}) \neq L(\vec{u})+L(\vec{v})$.

ex $L: \mathbb{R}_2 \to \mathbb{R}_2$. $L([x\ y]) = [x^2\ \ y]$. Is L a lin. trans? NO!

$L([1\ 0]) + L([2\ 0]) \overset{?}{=} L([3\ 0])$ Not a lin. trans.

$[1\ 0] + [4\ 0] = [5\ 0] \neq [9\ 0]$

ex $L: P_1 \to P_2$ $L[p(t)] = t\,p(t)$. Is L a lin. trans? Yes!

a) $L[p(t)+q(t)] = t(p(t)+q(t)) = tp(t)+tq(t) = L[p(t)]+L[q(t)]$ ✓

b) $L[cp(t)] = t(cp(t)) = c(tp(t)) = cL[p(t)]$ ✓

Thm $L: V \to W$ is a lin. trans. Then:

a) $L(\vec{0}_V) = \vec{0}_W$ proof. $L(\vec{0}_V) = L(\vec{0}_V+\vec{0}_V) = L(\vec{0}_V)+L(\vec{0}_V) = 2L(\vec{0}_V) \Rightarrow L(\vec{0}_V) = \vec{0}_W$

b) $L(\vec{u}-\vec{v}) = L(\vec{u})-L(\vec{v})$.

05/05

ex $L: \mathbb{R}^2 \to \mathbb{R}^3$. $L\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x+1 \\ y+x \\ yx \end{bmatrix}$. Is L a lin. trans.?

$L\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow L(O_{\mathbb{R}^2}) \neq O_{\mathbb{R}^3}$. L is not lin. trans.

ex $L: \mathbb{R}_2 \to \mathbb{R}_2$ lin. trans. $L([1\ 1]) = [1\ -2]$, $L([-1\ 1]) = [2\ \ 3]$.

(a) Find $L([-1\ \ 5])$

$[-1\ \ 5] = a[1\ 1] + b[-1\ \ 1] \Rightarrow \begin{matrix} a-b=-1 \\ a+b=5 \end{matrix} \Rightarrow \begin{matrix} a=2 \\ b=3 \end{matrix}$

$$L\left(\begin{bmatrix}-1 & 5\end{bmatrix}\right) = L\left(2\begin{bmatrix}1 & 1\end{bmatrix} + 3\begin{bmatrix}-1 & 1\end{bmatrix}\right) = L\left(2\begin{bmatrix}1 & 1\end{bmatrix}\right) + L\left(3\begin{bmatrix}-1 & 1\end{bmatrix}\right)$$

$$= 2L\left(\begin{bmatrix}1 & 1\end{bmatrix}\right) + 3L\left(\begin{bmatrix}-1 & 1\end{bmatrix}\right)$$

$$= 2\begin{bmatrix}1 & -2\end{bmatrix} + 3\begin{bmatrix}2 & 3\end{bmatrix} = \begin{bmatrix}8 & 5\end{bmatrix}.$$

(b) $L\left(\begin{bmatrix}x & y\end{bmatrix}\right) = ?$

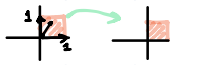
$$\begin{bmatrix}x & y\end{bmatrix} = a\begin{bmatrix}1 & 1\end{bmatrix} + b\begin{bmatrix}-1 & 1\end{bmatrix} \Rightarrow \begin{matrix}a-b=x \\ a+b=y\end{matrix} \Rightarrow \begin{matrix}a=(x+y)/2 \\ b=(y-x)/2\end{matrix}$$

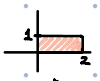
$$L\left(\begin{bmatrix}x & y\end{bmatrix}\right) = L\left(\frac{(x+y)}{2}\begin{bmatrix}1 & 1\end{bmatrix} + \frac{(y-x)}{2}\begin{bmatrix}-1 & 1\end{bmatrix}\right) = \frac{(x+y)}{2}L\left(\begin{bmatrix}1 & 1\end{bmatrix}\right) + \frac{(y-x)}{2}L\left(\begin{bmatrix}-1 & 1\end{bmatrix}\right)$$

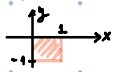
$$= \frac{(x+y)}{2}\begin{bmatrix}1 & -2\end{bmatrix} + \frac{(y-x)}{2}\begin{bmatrix}2 & 3\end{bmatrix} = \begin{bmatrix}\frac{x+y}{2} + \frac{2y-2x}{2} & \frac{-2x-2y+3y-3x}{2}\end{bmatrix} = \begin{bmatrix}\frac{-x+3y}{2} & \frac{-5x+y}{2}\end{bmatrix}$$

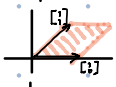
Verify $L\left(\begin{bmatrix}1 & 5\end{bmatrix}\right) = \begin{bmatrix}\frac{1+15}{2} & \frac{5+5}{2}\end{bmatrix} = \begin{bmatrix}8 & 5\end{bmatrix}$

St. Mat. Rep.

$L: \mathbb{R}^2 \to \mathbb{R}^2$. $L\left(\begin{bmatrix}x \\ y\end{bmatrix}\right) = \begin{bmatrix}x \\ y\end{bmatrix}$  $A = \begin{bmatrix}L\left(\begin{bmatrix}1 \\ 0\end{bmatrix}\right) & L\left(\begin{bmatrix}0 \\ 1\end{bmatrix}\right)\end{bmatrix} = \begin{bmatrix}1 & 0 \\ 0 & 1\end{bmatrix} = I_2$.

$L\left(\begin{bmatrix}x \\ y\end{bmatrix}\right) = \begin{bmatrix}2x \\ y\end{bmatrix}$  $A = \begin{bmatrix}2 & 0 \\ 0 & 1\end{bmatrix}$ $\det(A) = 2$

$L\left(\begin{bmatrix}x \\ y\end{bmatrix}\right) = \begin{bmatrix}x \\ -y\end{bmatrix}$  $A = \begin{bmatrix}1 & 0 \\ 0 & -1\end{bmatrix}$

$L\left(\begin{bmatrix}x \\ y\end{bmatrix}\right) = \begin{bmatrix}x+y \\ y\end{bmatrix}$  $A = \begin{bmatrix}1 & 1 \\ 0 & 1\end{bmatrix}$

$L\left(\begin{bmatrix}x \\ y\end{bmatrix}\right) = \begin{bmatrix}x \\ 0\end{bmatrix}$ $A = \begin{bmatrix}1 & 0 \\ 0 & 0\end{bmatrix}$

$L: \mathbb{R}^n \to \mathbb{R}^m$ lin. trans. $\{\vec{e}_1, \dots, \vec{e}_n\}$ standard basis. $\vec{e}_1 = \begin{bmatrix}1 \\ 0 \\ \vdots \\ 0\end{bmatrix}, \vec{e}_2 = \begin{bmatrix}0 \\ 1 \\ \vdots \\ 0\end{bmatrix}, \dots, \vec{e}_n = \begin{bmatrix}0 \\ 0 \\ \vdots \\ 1\end{bmatrix}_{n\times 1}$

Let A be the $m \times n$ matrix whose jth column is $L(e_j)$. Then $L(\vec{x}) = A\vec{x}$ $\forall x \in \mathbb{R}^n$.

A is called the **standard matrix representation** of L.

ex $L: \mathbb{R}^3 \to \mathbb{R}^2$. $L\left(\begin{bmatrix}x \\ y \\ z\end{bmatrix}\right) = \begin{bmatrix}x+2y \\ 3y-2z\end{bmatrix}$ Find the st. mat. rep.

$$A = \begin{bmatrix}L(e_1) & L(e_2) & L(e_3)\end{bmatrix} = \begin{bmatrix}1 & 2 & 0 \\ 0 & 3 & -2\end{bmatrix} \qquad e_1 = \begin{bmatrix}1 \\ 0 \\ 0\end{bmatrix}, e_2 = \begin{bmatrix}0 \\ 1 \\ 0\end{bmatrix}, e_3 = \begin{bmatrix}0 \\ 0 \\ 1\end{bmatrix}.$$

ex. $L: P_1 \to P_1$ lin. trans. $L(t+1) = 2t+3$, $L(t-1) = 3t-2$. Find $L(6t-4)$.

$$6t-4 = a(t+1) + b(t-1) \Rightarrow \begin{matrix}a+b=6 \\ a-b=-4\end{matrix} \Rightarrow \begin{matrix}a=1 \\ b=5\end{matrix}$$

$$L(6t-4) = L(1(t+1) + 5(t-1)) = L(t+1) + 5L(t-1) = 2t+3 + 5(3t-2) = 17t+1.$$

05/11

CHAPTER 7 / 7.1. Eigenvalues and eigenvectors

V = v. space. $L: V \to V$ lin. trans.

If there is a vector $\vec{x} \in V$, $\vec{x} \neq \vec{0}_V$ and $\lambda \in \mathbb{C}$ (complex number) such that $L\vec{x} = \lambda\vec{x}$

then $\vec{x}$ is called an **eigenvector** and λ is called an **eigenvalue** of L.

For us: $L = A_{n\times n}$ $\quad V = \mathbb{R}^n$. $\quad A\vec{x} = \lambda\vec{x}$

ex $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$. Find all eigenvalues and eigenvectors.

$$A\vec{x} = \lambda\vec{x} \iff \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \iff \begin{matrix} x_1 + x_2 = \lambda x_1 \\ -2x_1 + 4x_2 = \lambda x_2 \end{matrix} \iff \begin{matrix} (1-\lambda)x_1 + x_2 = 0 \\ -2x_1 + (4-\lambda)x_2 = 0 \end{matrix} \iff \begin{bmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The system has non-trivial solutions if and only if $\det\begin{pmatrix} 1-\lambda & 1 \\ -2 & 4-\lambda \end{pmatrix} = 0$ $\longrightarrow \det(A - \lambda I_2)$

$(1-\lambda)(4-\lambda) + 2 = 0 \iff 6 - 5\lambda + \lambda^2 = 0 \iff \lambda = 2,\ \lambda = 3$. Eigenvalues.

$\lambda = 2$: $\begin{bmatrix} 1-2 & 1 \\ -2 & 4-2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad \left[\begin{array}{cc|c} -1 & 1 & 0 \\ -2 & 2 & 0 \end{array}\right] \xrightarrow{-2r_1 + r_2 \to r_2} \left[\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$ $\quad -x_1 + x_2 = 0$ $\quad x_2 = r$, $x_1 = r$

Eigenvectors of $\lambda = 2$ are of the form $\begin{bmatrix} r \\ r \end{bmatrix}$, $r \neq 0$.

$\lambda = 3$: $\begin{bmatrix} 1-3 & 1 \\ -2 & 4-3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad \begin{bmatrix} -2 & 1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \iff -2x_1 + x_2 = 0$ $\quad x_1 = r$, $x_2 = 2r$

Eigenvectors of $\lambda = 3$ are of the form $\begin{bmatrix} r \\ 2r \end{bmatrix}$, $r \neq 0$.

def. $A_{n\times n}$ matrix. $\quad \det(cA)_{n\times n} = c^n \det(A)$

$p(\lambda) = \det(\lambda I - A)$ = **characteristic polynomial** of A.

$p(\lambda) = 0 \iff$ eigenvalues

characteristic equation.

ex $A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$. Find the char. poly. of A.

$p(\lambda) = \det(\lambda I_3 - A)$

$$\lambda I_3 - A = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix} = \begin{bmatrix} \lambda - 1 & -2 & 1 \\ -1 & \lambda & -1 \\ -4 & 4 & \lambda - 5 \end{bmatrix}$$

$$\det(\lambda I - A) = (-1)^{1+1}(\lambda - 1)\begin{vmatrix} \lambda & -1 \\ 4 & \lambda - 5 \end{vmatrix} + (-1)^{1+2}(-2)\begin{vmatrix} -1 & -1 \\ -4 & \lambda - 5 \end{vmatrix} + (-1)^{1+3}\begin{vmatrix} -1 & \lambda \\ -4 & 4 \end{vmatrix}$$

$$= (\lambda - 1)\left(\lambda(\lambda - 5) + 4\right) + 2\left(-(\lambda - 5) - 4\right) + \left(-4 + 4\lambda\right)$$

$$= \lambda^3 - 6\lambda^2 + 11\lambda - 6$$

Thm The roots of the char. poly $p(\lambda) = 0$ are the eigenvalues of the matrix A.

ex Find the eigenvalues and eigenvectors of A in last example.

eigenvalues are roots of $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3) = 0 \iff \lambda^3 - (\lambda_1 + \lambda_2 + \lambda_3)\lambda^2 + (\lambda_1\lambda_2 + \lambda_1\lambda_3 + \lambda_2\lambda_3)\lambda - \lambda_1\lambda_2\lambda_3 = 0$

$-\lambda_1\lambda_2\lambda_3 = -6 \iff$ any easy solutions?

$ax^2 + bx + c = 0 \Rightarrow x_{1,2} = \frac{-b \pm \sqrt{\Delta}}{2a}$

$ax^3 + bx^2 + cx + d = 0 \Rightarrow$ formulas

$ax^4 + \dots + e = 0 \Rightarrow$?

integer solutions $\pm1, \pm2, \pm3, \pm6$.

$\lambda=1 \Rightarrow 1-6+11-6=0$

$\lambda^3-6\lambda^2+11\lambda-6=(\lambda-1)\underbrace{(a\lambda^2+b\lambda+c)}_{\to 1}=a\lambda^3+\lambda^2(-a+b)+\lambda(c-b)-c \Rightarrow \begin{matrix} a=1 \\ b-a=-6 \Rightarrow b=-6+a=-5 \\ -c=-6 \Rightarrow c=6 \end{matrix}$

$p(\lambda)=(\lambda-1)(\lambda^2-5\lambda+6)=(\lambda-1)(\lambda-2)(\lambda-3)$.

Eigenvalues: $\lambda=1$, $\lambda=2$, $\lambda=3$.

Eigenvectors. $\lambda=1 \Rightarrow (1I_3-A)\vec{x}=\vec{0} \Rightarrow \begin{bmatrix} 1-1 & -2 & 1 \\ -1 & 1 & -1 \\ -4 & 4 & 1-5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|c} 0 & -2 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ -4 & 4 & -4 & 0 \end{array}\right] \begin{matrix} \\ \\ -r_2+r_3 \to r_3 \end{matrix}$

$\left[\begin{array}{ccc|c} 0 & -2 & 1 & 0 \\ -1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{matrix} r_1 \leftrightarrow r_2 \\ -r_1 \to r_1 \end{matrix} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{matrix} \\ -\frac{1}{2}r_2 \to r_2 \end{matrix} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \begin{matrix} x_3=r \\ x_2=r/2 \end{matrix} \quad \begin{matrix} x_1-x_2+x_3=0 \\ x_1-\frac{r}{2}+r=0 \Rightarrow x_1=-\frac{r}{2} \end{matrix}$.

Eigenvectors of $\lambda=1 \Rightarrow \begin{bmatrix} -r/2 \\ r/2 \\ r \end{bmatrix}=r\begin{bmatrix} -1/2 \\ 1/2 \\ 1 \end{bmatrix}$, $r\neq 0$

Another way to write: $s\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$, $s\neq 0$

Eigenvectors of $\lambda=2$, $\lambda=3$ can be found in the same way. Homework. Example 12, 7.1.

ex. $A=\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Find eigenvalues and eigenvectors.

1st step. Find the char. poly.

$\det(\lambda I_2-A)=p(\lambda)=\det\begin{pmatrix} \lambda-0 & +1 \\ -1 & \lambda-0 \end{pmatrix}=\lambda^2+1$

2nd step. $p(\lambda)=\lambda^2+1=0 \Leftrightarrow \lambda^2=-1 \Leftrightarrow \lambda_1=-\sqrt{-1}$ ($-i$), $\lambda_2=+\sqrt{-1}$ (i)

$i^2=-1$

$\lambda_1=-i$, $\lambda_2=i$.

3rd step. $\lambda_1=-i$ $\begin{pmatrix} -i-0 & +1 \\ -1 & -i-0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\left[\begin{array}{cc|c} -i & 1 & 0 \\ -1 & -i & 0 \end{array}\right] r_1 \leftrightarrow r_2 \left[\begin{array}{cc|c} -1 & -i & 0 \\ -i & 1 & 0 \end{array}\right] -ir_1+r_2 \to r_2 \left[\begin{array}{cc|c} -1 & -i & 0 \\ -i+i & 1-1 & 0 \end{array}\right]$

$-x_1-ix_2=0$ $\begin{matrix} x_2=r \\ x_1=-ir \end{matrix} \Rightarrow \vec{x}=\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}=\begin{bmatrix} -ir \\ r \end{bmatrix}=r\begin{bmatrix} -i \\ 1 \end{bmatrix}$, $r\neq 0$ → eigenvectors.

$\lambda_2=i$ .... Carry on the operations. $\vec{x}=r\begin{bmatrix} i \\ 1 \end{bmatrix}$, $r\neq 0$ ↳ can be a complex number.

ex $L: P_2 \to P_2$, $L(at^2+bt+c)=-bt-2c$. lin. trans.

(a) Find the matrix representation of L with respect to the basis $S=\{\underbrace{1-t}_{\vec{v}_1}, \underbrace{1+t}_{\vec{v}_2}, \underbrace{t^2}_{\vec{v}_3}\}$.

$L(\vec{v}_1)=a_1\vec{v}_1+a_2\vec{v}_2+a_3\vec{v}_3$

$L(1-t)=t-2=a_1(1-t)+a_2(1+t)+a_3t^2$ $\begin{matrix} a_3=0 \\ a_2-a_1=1 \\ a_1+a_2=-2 \end{matrix}$ $2a_2=-1$ $a_2=-1/2$ $a_1=-3/2$

$[L(1-t)]_S=\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}=\begin{bmatrix} -3/2 \\ -1/2 \\ 0 \end{bmatrix}$

$[L(1+t)]_S=\begin{bmatrix} -1/2 \\ -3/2 \\ 0 \end{bmatrix}$

$[L(t^2)]_S=\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

The mat. rep. of L w.r.t. ordered basis S is $\begin{bmatrix} -3/2 & -1/2 & 0 \\ -1/2 & -3/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} = A$

(b) Find the eigenvalues and eigenvectors of A.

12/05/2020

7.2. Diagonalization.

$A_{2\times 2}$ $\quad A\vec{v}_1 = \lambda_1\vec{v}_1$ $\quad A\vec{v}_2 = \lambda_2\vec{v}_2$ $\quad \lambda_1 \neq \lambda_2$. $\quad \vec{v}_1, \vec{v}_2$: eigenvectors.

$P = [\vec{v}_1 \ \vec{v}_2]_{2\times 2}$ $\quad AP = A[\vec{v}_1 \ \vec{v}_2] = [A\vec{v}_1 \ A\vec{v}_2]$

proof. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} = \left[\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \ \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right]$ first col: $\begin{bmatrix} ax_1 + bx_2 \\ cx_2 + dx_2 \end{bmatrix}$

$= [\lambda_1\vec{v}_1 \ \lambda_2\vec{v}_2]$ ✓ proved!

$= [\vec{v}_1 \ \vec{v}_2]\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = P\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

If P^{-1} exists $\Rightarrow P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$ = a diagonal matrix $= D$

$P^{-1}AP = D \Leftrightarrow PP^{-1}AP = PD \Leftrightarrow APP^{-1} = PDP^{-1} \Leftrightarrow A = PDP^{-1}$

Why is this useful?

$A^2 = AA = (PDP^{-1})(PDP^{-1}) = PDDP^{-1} = PD^2P^{-1}$

$A^n = PD^nP^{-1}$ (HARD $\quad$ EASY)

$D^2 = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{bmatrix}$

$D^n = \begin{bmatrix} \lambda_1^n & 0 \\ 0 & \lambda_2^n \end{bmatrix}$

DEF $A_{n\times n}$. If P^{-1} exists and D is diagonal mat. and $A = PDP^{-1}$ then A is called diagonalizable.

OPTIONAL

Why is computing A^n important?

$e^A \overset{\text{def}}{=} \sum_{n=0}^{\infty} \frac{A^n}{n!}$

Calculus 1: $\frac{d}{dx}e^{ax} = ae^{ax}$

Advanced lin. alg. $\frac{d}{dt}e^{At} = Ae^{At}$. This means: $\underbrace{\frac{d}{dt}\vec{x}(t) = A\vec{x}(t), \ \vec{x}(0) = \vec{x}_0}_{\text{INITIAL VALUE PROBLEM}} \Rightarrow \underbrace{\vec{x}(t) = e^{At}\vec{x}_0}_{\text{SOLUTION}}$

When does P^{-1} exist?

It exists if $\vec{v}_1, \vec{v}_2$ are eigenvectors of distinct eigenvalues.

ex $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ Is A diagonalizble, if so find P, D (diagonal matrix) such that $A = PDP^{-1}$.

Previously we found: $\lambda_1 = 2$ $\vec{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\lambda_2 = 3$, $\vec{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1}$

Check: $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}^{-1} = \frac{1}{1\cdot 2 - 1\cdot 1}\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$ H.W. Check it!

Thm (a) If $A_{n\times n}$ matrix. Suppose $\{\vec{v}_1, \ldots, \vec{v}_k\}$ are eigenvectors belonging to

distinct eigenvalues. Then $\{\vec{v}_1, \ldots, \vec{v}_k\}$ are lin. indep.

(b) If A has n distinct eigenvalues then A is diagonalizable.

(c) A has n lin. indep. eigenvectors if and only if then A is diagonalizable.

ex $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Is A diagonalizable?

$p(\lambda) = \det(\lambda I - A) = \det\begin{pmatrix} \lambda-1 & -1 \\ 0 & \lambda-1 \end{pmatrix} = (\lambda-1)^2$ = char. poly.

$\lambda = 1$ is the only eigenvalue.

Find the eigenvectors. $\begin{pmatrix} 1-1 & -1 \\ 0 & 1-1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{matrix} -x_2 = 0 \\ 0 = 0 \end{matrix} \Rightarrow \begin{matrix} x_2 = 0 \\ x_1 = r \end{matrix}$, $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = r\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ ← only 1 indep. eigen vector

NOT DIAGONALIZABLE

Optional. 4.9. Rank of a matrix.

$A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 1 & 4 \end{bmatrix}$ Number of independent rows = Number of independent cols. = 2 = rank of a matrix.

$A_{m \times n}$ matrix. $A\vec{x} = \vec{0}$. Sol. set = nullspace of A (subspace of $\mathbb{R}^n$)

Fundamental Theorem of Lin. Algebra: rank_A of + (dim. of nullspace of A) = number of columns of A.

↳ nullity

ex $A = \begin{bmatrix} 1 & 2 & 5 \\ 3 & 1 & 4 \end{bmatrix}$ rank = 2 $\overset{\text{FTLA}}{\Rightarrow}$ nullity = 1

$A\vec{x} = \vec{0}$ has 1 lin. indep. sol.

CHAPTER 1

1.4

37. The linear system $ACx = b$ is such that A and C are nonsingular with

$$A^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}, \quad C^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \text{ and } b = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Find all 2×2 matrices with real entries of the form

$$A = \begin{bmatrix} a & b \\ 0 & c \end{bmatrix}$$

such that $A^2 = I_2$.

Chapter 1 Supplementary Exercises

10. Show that the product of two 2×2 skew symmetric matrices is diagonal. Is this true for $n \times n$ skew symmetric matrices with $n > 2$?

Prove that if $\text{Tr}(A^T A) = 0$, then $A = O$.

CHAPTER 2

2.2

11. Find a 2×1 matrix $\mathbf{x}$ with entries not all zero such that

$$A\mathbf{x} = 3\mathbf{x}, \quad \text{where } A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

14. In the following linear system, determine all values of a for which the resulting linear system has

(a) no solution;

(b) a unique solution;

(c) infinitely many solutions:

$$\begin{aligned} x + y - z &= 2 \\ x + 2y + z &= 3 \\ x + y + (a^2 - 5)z &= a \end{aligned}$$

17. Repeat Exercise 14 for the linear system

$$\begin{aligned} x + y &= 3 \\ x + (a^2 - 8)y &= a. \end{aligned}$$

26. Find an equation relating a, b, and c so that the linear system

$$\begin{aligned} x + 2y - 3z &= a \\ 2x + 3y + 3z &= b \\ 5x + 9y - 6z &= c \end{aligned}$$

is consistent for any values of a, b, and c that satisfy that equation.

In the following linear system, determine all values of a for which the resulting linear system has

(a) no solution;

(b) a unique solution;

(c) infinitely many solutions:

$$\begin{aligned} x + y - z &= 2 \\ x + 2y + z &= 3 \\ x + y + (a^2 - 5)z &= a \end{aligned}$$

2.3

Using elementary matrices.

7. Find the inverse of $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$.

17. Which of the following homogeneous systems have a nontrivial solution?

(a)
$$\begin{aligned} x + 2y + 3z &= 0 \\ 2y + 2z &= 0 \\ x + 2y + 3z &= 0 \end{aligned}$$

(b)
$$\begin{aligned} 2x + y - z &= 0 \\ x - 2y - 3z &= 0 \\ -3x - y + 2z &= 0 \end{aligned}$$

(c)
$$\begin{aligned} 3x + y + 3z &= 0 \\ -2x + 2y - 4z &= 0 \\ 2x - 3y + 5z &= 0 \end{aligned}$$

19. Find all value(s) of a for which the inverse of

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 2 & a \end{bmatrix}$$

exists. What is A^{-1}?

7. Find the inverse of

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

3. Let B be the matrix obtained from A after the row operations $2\mathbf{r}_3 \to \mathbf{r}_3$, $\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$, $4\mathbf{r}_1 + \mathbf{r}_3 \to \mathbf{r}_3$, and $-2\mathbf{r}_1 + \mathbf{r}_4 \to \mathbf{r}_4$ have been performed. If $\det(B) = 2$, find $\det(A)$.

4. Compute the determinant of

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 3 \\ -2 & 0 & -1 & -1 \\ 3 & 0 & 0 & -1 \end{bmatrix}$$

by using row operations to obtain upper triangular form.

7. Use the adjoint to compute A^{-1} for

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 1 & 1 \\ 2 & -1 & -2 \end{bmatrix}.$$

8. Solve the linear system $A\mathbf{x} = \mathbf{b}$ by using Cramer's rule, given

$$A = \begin{bmatrix} 3 & 4 & 2 \\ 1 & 2 & 2 \\ 3 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}.$$